

A complete rectangular function space without the subsequence property

A counterexample to the question in Section 4.2 of arXiv:1810.05714

Autonomous proof attempt, lane 6

13 August 2026

Status: candidate full counterexample, likely valid. A band-preserving involution of $L^0[0, 1]$ transports the L^2 norm to a complete strictly rectangular space whose norm topology is discontinuously embedded in convergence in measure.

1 The source question

Throughout, functions are real equivalence classes modulo null sets. A normed space $X(\mu) \subset L^0(\mu)$ is *rectangular* if there is C such that

$$f \in X, A \in \Sigma \implies \mathbf{1}_A f \in X, \quad \|\mathbf{1}_A f\|_X \leq C \|f\|_X.$$

It has the *subsequence property* if every norm-convergent sequence has a subsequence converging pointwise almost everywhere to the same limit.

Calabuig, Fernández Unzueta, Galaz-Fontes, and Sánchez Pérez ask at the start of Section 4.2:

We do not know if an arbitrary Banach rectangular f.s. has the subsequence property.

They prove the assertion for rectangular sequence spaces in Proposition 4.5 [1, Section 4.2]. The source passage, on PDF page 12, is reproduced in Figure 1.

2 Proof intuition

Rectangularity only sees multiplication by characteristic functions. On the nonatomic space $[0, 1]$, that Boolean action does not determine the usual measurable realization of a Banach space continuously. Local Hamel bases of the universally complete lattice L^0 allow a discontinuous real-linear change of coordinates which nevertheless commutes with every characteristic projection.

Choose such a coordinate involution U which swaps the constant function $\mathbf{1}$ with $q(t) = t$. The image $U(L^2)$, equipped with the transported L^2 norm, is automatically complete. Since U commutes with restrictions, the norm is even strictly rectangular. But on simple functions s one has $U(s) = qs$, whereas $U(q) = \mathbf{1}$. Approximating q by simple functions proves that $U : L^2 \rightarrow L^0$ is discontinuous. A norm-null sequence can therefore stay uniformly away from zero in measure, which prevents almost-everywhere convergence along every subsequence.

3 The local coordinate involution

Let

$$(\Omega, \Sigma, \mu) = ([0, 1], \mathcal{L}, \mathcal{L}^1), \quad G = L^0(\mu), \quad q(t) = t.$$

The vector lattice G is universally complete. Its band projections are exactly

$$P_A f = \mathbf{1}_A f \quad (A \in \Sigma),$$

where sets are understood modulo null sets.

We use the local Hamel basis theorem for universally complete vector lattices from Gutman, Kusraev, and Kutateladze [2, Sections 1.1–1.2]. A locally linearly independent set extends to a local Hamel basis $\mathcal{E}$. Every $f \in G$ then has, on a partition of unity, a finite linear expansion in the restrictions of elements of $\mathcal{E}$, and the local coefficients are unique [2, Sections 1.2.1–1.2.3].

Lemma 1. *There is a real-linear involution $U : G \rightarrow G$ such that*

$$U(\mathbf{1}) = q, \quad U(q) = \mathbf{1}, \quad UP_A = P_A U \quad (A \in \Sigma).$$

Proof. First, $\{\mathbf{1}, q\}$ is locally linearly independent. Indeed, if A has positive measure and

$$\mathbf{1}_A(a\mathbf{1} + bq) = 0,$$

then $a + bt = 0$ for almost every $t \in A$. If $b \neq 0$, this would put A modulo a null set inside the singleton $\{-a/b\}$, which is impossible. Thus $b = 0$ and then $a = 0$.

Extend $\{\mathbf{1}, q\}$ to a local Hamel basis $\mathcal{E}$ of G . Let $\sigma : \mathcal{E} \rightarrow \mathcal{E}$ swap $\mathbf{1}$ and q and fix all other basis elements. For $f \in G$, choose a local representation

$$f = \sum_{\xi} \mathbf{1}_{A_{\xi}} \left(\sum_{e \in F_{\xi}} \lambda_{\xi, e} e \right), \quad (1)$$

where (A_{ξ}) is a measurable partition modulo null sets and every $F_{\xi} \subset \mathcal{E}$ is finite. Define

$$Uf = \sum_{\xi} \mathbf{1}_{A_{\xi}} \left(\sum_{e \in F_{\xi}} \lambda_{\xi, e} \sigma(e) \right). \quad (2)$$

The uniqueness of local coefficients shows, after passing to a common refinement, that (2) is independent of (1). The same common-refinement argument gives linearity. Since σ^2 is the identity, so is U^2 . Finally, restricting (1) to a measurable set before or after applying (2) gives the same local expansion. Hence $UP_A = P_A U$ for every A . The two displayed values of U follow from the definition of σ . $\square$

4 The counterexample

Define the linear subspace

$$X := U(L^2[0, 1]) \subset L^0[0, 1]$$

and, using the bijectivity of U , define

$$\|x\|_X := \|U^{-1}x\|_2 \quad (x \in X).$$

Theorem 2. *X is a Banach space of measurable functions with a strictly rectangular norm, but X does not have the subsequence property. Consequently the question in Section 4.2 of [1] has a negative answer.*

Proof. The map

$$U : L^2[0, 1] \longrightarrow X$$

is an isometric linear bijection by construction, so X is complete. For $f \in L^2$ and $A \in \Sigma$, Lemma 1 gives

$$P_A U(f) = U(P_A f) \in X$$

and

$$\|P_A U(f)\|_X = \|P_A f\|_2 \leq \|f\|_2 = \|U(f)\|_X. \quad (3)$$

Thus X is strictly rectangular.

We next show that the linear map

$$T := U|_{L^2} : L^2[0, 1] \longrightarrow L^0[0, 1], \quad (4)$$

where L^0 has the topology of convergence in measure, is discontinuous. For every simple function $s = \sum_{j=1}^m c_j \mathbf{1}_{A_j}$, the commutation relation in Lemma 1 yields

$$U(s) = \sum_{j=1}^m c_j P_{A_j} U(\mathbf{1}) = \sum_{j=1}^m c_j \mathbf{1}_{A_j} q = sq. \quad (5)$$

Choose simple functions $s_n \rightarrow q$ in L^2 . By (5), $U(s_n) = s_n q \rightarrow q^2$ in L^2 , hence in measure. If (4) were continuous, the same sequence would converge in measure to $U(q) = \mathbf{1}$. Uniqueness of limits in L^0 would give $q^2 = \mathbf{1}$ almost everywhere, a contradiction.

Since (4) is not continuous at zero and the sets

$$V_{\varepsilon, \delta} := \{g \in L^0 : \mu(|g| > \varepsilon) < \delta\}$$

form a neighborhood basis at zero, there are fixed $\varepsilon, \delta > 0$ and functions $f_n \in L^2$ such that

$$\|f_n\|_2 < \frac{1}{n}, \quad \mu(|U(f_n)| > \varepsilon) \geq \delta \quad (n \geq 1). \quad (6)$$

Put $x_n = U(f_n) \in X$. Then $\|x_n\|_X = \|f_n\|_2 \rightarrow 0$. No subsequence of (x_n) converges almost everywhere to zero: on a finite measure space such convergence would imply convergence in measure, contradicting the uniform lower bound in (6). Hence X fails the subsequence property. $\square$

Corollary 3. *The positive cone $X_+ = \{x \in X : x \geq 0 \text{ a.e.}\}$ is not closed.*

Proof. For Banach rectangular function spaces, source Lemma 2.8 says that closedness of the positive cone is equivalent to the subsequence property [1, Lemma 2.8]. Apply Theorem 2. $\square$

5 Verification, provenance, and limitations

The only nonconstructive ingredient is the extension of $\{\mathbf{1}, q\}$ to a local Hamel basis. After that extension, the involution is given explicitly on local coordinates, and all norm estimates are identities. The example is real-linear, uses the finite Lebesgue measure space required by the source, and satisfies rectangularity with the best constant $C = 1$.

The older Wickstead paper supplies the local basis representation, but does not state this transported- L^2 construction or connect it to the later rectangular-space question. A bounded novelty search covered the run indexes, the exact sentence and title from the source, and combinations of the terms *rectangular function space*, *subsequence property*, and *band preserving*. It found no later claimed resolution. This supports plausible novelty but is not a definitive certification.

The construction is choice-dependent and gives no concrete formula for U on arbitrary measurable functions. It does not affect the positive sequence space result in [1]; the nonatomic measure algebra is essential to the local coordinate pathology.

Human-review recommendation. Review as a likely valid full counterexample. The decisive audit is Lemma 1: verify that permutation of a local Hamel basis defines an everywhere-defined band-preserving linear involution. Once this is accepted, (3)–(6) are elementary.

ii) Let $X(\mu)_j$, $j = 1, 2$, be the B.f.s. associated to the Banach rectangular f.s. $X(\mu)$ with the norm $\|\cdot\|_j$. Then the map $I : X(\mu)_1 \rightarrow X(\mu)_2$ given by $I(f) := f$ is a linear isomorphism and both I and its inverse map are positive operators. It follows from i) that both are bounded and this corresponds to the norms $\|\cdot\|_1$ and $\|\cdot\|_2$ being equivalent. $\square$

4.2. Normed sequence spaces.

We do not know if an arbitrary Banach rectangular f.s. has the subsequence property. However, we will now show this holds for a rectangular sequence space.

We will say that $X(\mu) = (\Omega, \Sigma, \mu)$ is a *normed sequence space* (*Banach sequence space* when complete) when $\Omega := \mathbb{N}$, $\Sigma := 2^\Omega$ and μ is a finite measure such that $\mu(\{n\}) > 0, \forall n \in \mathbb{N}$. The specific measure will not be important. For definiteness we can take, for example, the measure defined by $\mu(A) := \sum_{n \in A} \frac{1}{2^n}$. Then any function $f : \mathbb{N} \rightarrow \mathbb{R}$ is measurable and, since the empty set is the only set in Σ with measure zero, each equivalence class in $L^0(\mu)$ has only one member. So in this context we will identify $L^0(\mu)$ with $L^0(\Sigma)$ and also this last space with the set of real sequences $\mathcal{S}$. Let $A \subset \mathbb{N}$, then considering the above identification, notice that $\chi_A := \{a_n\}$, where $a_n = 1$ when $n \in A$ and $a_n = 0$ when $n \notin A$.

Let $X(\mu)$ be a normed sequence space. For each $n \in \mathbb{N}$, we define the linear function $\pi_n : X(\mu) \rightarrow \mathbb{R}$ by $\pi_n(f) := f(n)$.

Figure 1: The open statement at the start of Section 4.2, PDF page 12 of arXiv:1810.05714.

References

- [1] J. M. Calabuig, M. Fernández Unzueta, F. Galaz-Fontes, and E. A. Sánchez Pérez, *Equivalent Norms in a Banach Function Space and the Subsequence Property*, arXiv:1810.05714 (2018).
- [2] A. E. Gutman, A. G. Kusraev, and S. S. Kutateladze, *The Wickstead Problem*, arXiv:0712.2378 (2007).